

Md. Repon Islam

Contact Information

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Education

Year	Degree	Institution	Details
2025	M.Sc. Engg. (CSE)	Khulna University of Engineering & Technology (KUET), Bangladesh	CGPA: 4.00/4.00
2022	B.Sc. Engg. (CSE)	Khulna University of Engineering & Technology (KUET), Bangladesh	CGPA: 3.93/4.00 (2nd position among 120 students)
2016	HSC (Science)	Dinajpur Board	GPA: 5.00/5.00
2014	SSC (Science)	Dinajpur Board	GPA: 5.00/5.00

Research Interests

- Artificial Intelligence and Machine Learning
- Deep Learning & Neural Networks
- Computer Vision and Image Processing
- Embedded Systems & IoT for Assistive Technologies
- Bio-informatics and Health Informatics

Academic Appointments

- **Assistant Professor, Department of Computer Science & Engineering, KUET (Jan 2026 – Present)**
- **Lecturer (Full-time), Department of Computer Science & Engineering, KUET (Dec 2022 – Dec 2025)**
- **Lecturer (Part-time), Department of Computer Science & Engineering, KUET (May 2022 – Dec 2022)**

Industry Experience

- **Technical advisor for several large scale government and private AI projects.**
- **AI Data Annotator (Audio Data Annotation), Intelligent Machines Limited, Bangladesh — Contract-based**
- **Android Developer (Java), Protttoy EdTech (Digital Education Platform) — Jun 2020 – Sep 2020**
- **Android (Kotlin) & Cloud Function Developer, Captain Earth (Real-life task-based well-being gaming platform) — Sep 2020 – Dec 2020**

Research Projects & Contributions

- *Multi-modal Smart Wheelchair System* — Developed a reliable, low-cost IoT-enabled smart wheelchair incorporating finger-gesture, hand-gesture and voice control; includes fall and obstacle detection using deep learning and sensors.
- *KUETian* — Highly used Android student companion app featuring profile management, automated class routine, library management and more for KUET students.
- *FitSim* — Android application for simulating and generating realistic fitness data for Google Fit integration.
- *MedScan* — Real-time medicine recognition app using AI on Android for pharmacy assistance.
- *Human-Computer Interaction by Hand Gesture* — 2D CNN-based system enabling touchless computer interaction through hand gestures.
- *Face Recognition Based Smart Security Locker* — Raspberry-Pi based smart locker employing face recognition using Haar-Cascade classifiers.
- *PI Weather Station* — Remote weather data acquisition system using Raspberry Pi and sensors with automatic data upload via cellular network.
- *My Group Feed* — Automated Facebook Group posts retrieval system with a custom UI for distraction-free reading.
- *Android SMS Gateway* — Android-based low-cost global SMS server enabling dual SIM control.
- *Pacman AI Game* — Multi-opponent AI game using A* search algorithms implemented in Python.
- *3D Environment Design* — Dynamic 3D model of a multi-storey building with full interior using OpenGL and C++.

Recent AI & Industry Projects

- *AI based Digital Gas Metering & Monitoring System* — Technical advisor for Implementing a nationwide GPS-based gas metering platform featuring IoT sensors, edge computing, artificial intelligence and LLM integration for real-time industrial gas monitoring, predictive analytics and automated billing.
- *Large-Scale OCR & Document Automation System* — Designed a deep learning and large language model pipeline to digitise and extract structured information from

millions of paper documents, streamlining government services and enterprise workflows.

- *Edge AI & Industrial IoT Solutions* — Deployed embedded AI models on microcontrollers and edge devices for occupancy detection, fire safety and hazard prediction in industrial and building environments, integrating data streams via MQTT and RESTful APIs.
- *Multimodal Sign-Language & Vision Assistive Systems* — Developing a sign-language translation framework and generative reading devices leveraging neural networks and large language models to assist visually impaired and hard-of-hearing users.

Selected Publications

1. **M. S. Sadi, M. Alotaibi, Md. Repon Islam, M. S. Islam, T. Alhmiedat & Z. Bassfar**, “Finger-Gesture Controlled Wheelchair with Enabling IoT Sensors,” *Sensors* (MDPI), vol. 22, p. 8716, 2022.
2. **M. M. Islam, O. Tayan, Md. Repon Islam, M. S. Islam, S. Nooruddin, M. N. Kabir & M. R. Islam**, “Deep Learning Based Systems Developed for Fall Detection: A Review,” *IEEE Access*, vol. 8, pp. 166117–166137, 2020 (Impact Factor: 3.745, Q1).
3. **M. M. Islam, Md. Repon Islam & M. S. Islam**, “An Efficient Human Computer Interaction through Hand Gesture Using Deep Convolutional Neural Network,” *SN Computer Science*, Springer, vol. 1, issue 4, article 211, 2020.
4. **M. M. Islam, N. H. Neom, M. S. Imtiaz, S. Nooruddin, Md. Repon Islam & M. R. Islam**, “A Review on Fall Detection Systems Using Data from Smartphone Sensors,” *Ingénierie des Systèmes d’Information* (Information Systems Engineering), Lavoisier, vol. 24, issue 6, pp. 569–576, 2019.
5. **Md. Repon Islam & M. S. Sadi**, “Adapting Open-source Multimodal Large Language Models for Enhanced Vision Assistants,” in *Proc. 27th Int’l Conf. on Computer and Information Technology (ICCIT)*, pp. 3560–3565, IEEE, 2024.
6. **Md. Repon Islam & M. S. Sadi**, “Generative AI-assisted Standalone Wearable Reading Glasses for the Visually Impaired,” in *Proc. IEEE 3rd Int’l Conf. on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, pp. 173–176, IEEE, 2024 — *Best Paper Award*.
7. **G. T. Roy, A. I. Atik, M. R. Rabby & Md. Repon Islam**, “Enhancing Fire Safety: Encoder-Decoder Based Detection and IoT-Driven Response System,” in *Proc. 2nd Int’l Conf. on Information and Communication Technology (ICICT)*, pp. 219–223, IEEE, 2024.
8. **H. Neha, M. S. Sadi, Md. Repon Islam & S. Sakib**, “Automated Object Identification for the Visually Impaired,” in *Proc. 27th Int’l Conf. on Computer and Information Technology (ICCIT)*, pp. 3584–3589, IEEE, 2024.
9. **S. Sakib, M. S. Sadi, M. S. Arefin, Md. Repon Islam & H. Neha**, “An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading,” in *Proc. 13th Int’l Conf. on Electrical and Computer Engineering (ICECE)*, pp. 767–772, IEEE, 2024.

10. **Md. Repon Islam, M. S. Islam & M. S. Sadi**, “Towards Developing a Real-Time Hand Gesture Controlled Wheelchair,” in *Proc. Int’l Conf. on Trends in Computational and Cognitive Engineering (TCCE)*, pp. 355–366, Springer, 2020.

Awards & Honors

- **Best Paper Award**, IEEE RAAICON 2024 for the paper “Generative AI-assisted Standalone Wearable Reading Glasses for the Visually Impaired.”
- **Second position** (2nd out of 120 students) in B.Sc. Engineering (CSE), KUET, 2022.

Technical Skills

- Programming Languages: Assembly, C, C++, Python, Java, Swift, Dart
- Machine Learning & Deep Learning: TensorFlow, PyTorch, Keras, TensorFlow Lite, ONNX
- Computer Vision & NLP: OpenCV, scikit-learn, Large Language Models (LLM) integration
- Edge & Embedded Systems: Raspberry Pi, NVIDIA Jetson, Microcontrollers; IoT protocols (MQTT, HTTP)
- Web Technologies: PHP, ASP.NET, JavaScript, HTML/CSS, Node.js
- Databases: MySQL, SQL Server, Oracle
- Cloud & DevOps: AWS, Azure, Docker, Kubernetes, RESTful APIs, CI/CD
- Tools & Platforms: Android Studio, Xcode, Anaconda, CISCO Packet Tracer
- Version Control: Git
- Operating Systems: Windows, Linux-based OS, macOS, Android, iOS, Raspbian

Referees

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